



Learning Python - Sheet #3

Loops

Loops are the last important concept we need before we can do some serious programming. We'll see 2 kinds of loops: **for loops** and **while loops**.

1. For loops

A for loop allows to do something for each object in a series of objects (called a **sequence**).

Try this:

```
for name in ["Richard", "Elizabeth", "Guido"]:  
    print("Hello", name)
```

You can see that as for functions, shows "..." instead of ">>>" and automatically adds **indentation** when it's waiting for your **for loop body**.

The square brackets allowed us to define a **list** which contains 3 **strings** (the names). You can try replacing it with an empty list. You can define an empty list with **[]** (which just means removing all the names).

There's a special function called **range** which allows to generate a sequence of numbers (**integers**). Try the following:

```
for i in range(5):  
    print(i)
```



Pay attention:

Note how it started at 0 and it didn't include 5. Computer scientists love starting at 0 and stopping before reaching the upper boundary!

Have fun experimenting:

You can also call **range** with more parameters. Can you understand what they do?

```
for i in range(5, 10):  
    print(i)  
  
for i in range(2, 20, 3):  
    print(i)
```

2. While loops

A while loop allows to repeat some action until some condition is met. Try this:

```
answer = ""  
while answer != "yes":  
    answer = input("Do you want to quit this loop? ")
```



Pay attention:

Note how we have to create (**define**) the variable **answer** before the loop. That's because if we don't, it doesn't exist the first time the condition is **evaluated** to decide whether we enter the loop.



Have fun experimenting:

Can you quit the loop without answering yes? Can you answer yes without quitting the loop?

Here is a way to write the same loop without having to define **answer** before. Do you understand how it works?

```
while True:  
    answer = input("Do you want to quit this loop? ")  
    if answer == "yes":  
        break
```

